VISHNU INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

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HealthMate: Modern BMI & Health Calculator

A Responsive Flutter Web Mobile Application with Material 3 & GitHub Actions CI/CD

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1. Executive Summary & Project Description

HealthMate is a production-grade, responsive health and fitness application engineered using the **Flutter** framework and **Dart** language. Built strictly following **Material 3** design principles, the system operates as a zero-backend, client-side Progressive Web Application (PWA) that computes essential health metrics entirely within the user's browser, ensuring absolute user privacy with no remote database, authentication, or external API overhead.

In modern lifestyles, individuals encounter fragmented fitness utilities and intrusive trackers requiring mandatory logins. HealthMate addresses this by delivering a lightweight, offline-ready dashboard that unifies four vital biometric estimators: **Body Mass Index (BMI)** with interactive visual categorization, **Daily Water Intake** with real-time glass tracking, **Total Daily Energy Expenditure (TDEE/BMR)** through the clinical Mifflin-St Jeor formula, and **Ideal Body Weight** cross-analyzed through the Devine, Robinson, and WHO frameworks. The application is deployed via automated continuous integration and continuous deployment (CI/CD) pipelines using GitHub Actions directly to GitHub Pages.

2. Core Application Modules Overview

Module	Underlying Formulation / Scientific Standard	Key Interactive Features
BMI Calculator	Quetelet Index: Weight (kg) / [Height (m)]² WHO 4-Tier classification: <18.5, 18.5-24.9, 25-29.9, ≥30	Gender selection cards, Age validation, Dual unit toggles (cm/ft, kg/lb), Visual 4-color animated gauge, Target range.
Water Intake	Baseline hydration (~33 mL/kg) + climate allowance (+500 mL) + exertion factor (+400/800 mL)	Interactive logger (+250 mL glass, -250 mL), animated circular progress meter, hydration status percentage.
Calorie Needs	Mifflin-St Jeor BMR equation + 5-tier Activity Multipliers (1.2 to 1.9) + Goal offsets (-500 to +300 kcal)	Basal Metabolic Rate vs Maintenance TDEE, Balanced macronutrient split (30% Protein, 40% Carbs, 30% Fat).
Ideal Weight	Devine (1974), Robinson (1983) and WHO Healthy BMI (18.5–24.9) bounds	Triple formula comparison matrix, current weight differential check, frame-specific wellness recommendations.
Health Insights	Evidence-based lifestyle practices across Hydration, Nutrition, Sleep, and Exercise	Interactive category filtering chips, read-time chips, and expandable modal bottom sheets with full advice.
User Profile	Local state management for client preferences	Displaying user **Ganesh** (ganesh@gmail.com), Metric/Imperial switch, Light/Dark appearance toggle.

3. System Architecture & Technical Design

HealthMate follows a decoupled layered architecture separating presentation widgets, business computation models, custom theme tokens, and cloud deployment pipelines. All computations execute synchronously within the client JavaScript runtime produced by the Flutter Web compilation engine.

Architectural Tier	Components & Implementation	Responsibilities & Design Patterns
1. Presentation Layer (UI)	MainNavigationShell, HomeDashboardScreen, BMICalculatorScreen, WaterCalculatorScreen	Material 3 UI, declarative widgets, responsive mobile-to-desktop layout constraints (maxWidth: 860px), smooth tab switching via IndexedStack.
2. Custom Widget Library	BMIScaleBar, WaterMeter, CustomCard, UnitToggle, DisclaimerCard	Atomic design components, reusable elevation shadows, custom canvas gauges, animated progress indicators, segmented unit pills.
3. Calculation Engine (Models)	BMIResult, CalorieResult, WaterIntakeResult, IdealWeightResult	Immutable state models with pure static evaluation functions. Converts imperial to metric, performs boundary validation, and categorizes status.
4. Theme & Styling Layer	AppTheme (Light/Dark themes), CardThemeData, Google Fonts (Plus Jakarta Sans)	Dynamic ThemeMode toggling, emerald green (#10B981) and ocean blue (#2563EB) accents, standardized text hierarchies and borders.
5. CI/CD Pipeline Layer	GitHub Actions (deploy.yml), subosito/flutter-action@v2, actions/deploy-pages@v4	Cloud Ubuntu runners compile Web release binaries, calculate dynamic base-href (/BMI/), and publish to GitHub Pages.

4. Project Structure & Directory Layout

```
BMI/ (HealthMate Project Root)
├── .github/workflows/deploy.yml         [Cloud CI/CD: Flutter setup, analyze, build & GitHub Pages deploy]
├── analysis_options.yaml                 [Flutter & Dart lint analysis rules with no fatal warnings]
├── pubspec.yaml                             [Project metadata, Google Fonts dependency, Material 3 flags]
├── README.md                                [Detailed project guide, deployment URL, browser upload steps]
├── web/
│   ├── favicon.svg                            [SVG heart pulse emblem in emerald green & ocean blue]
│   ├── index.html                             [HTML5 container with dynamic $FLUTTER_BASE_HREF support]
│   └── manifest.json                          [Progressive Web App manifest for standalone browser launch]
└── lib/
    ├── main.dart                            [App entry point, stateful ThemeMode toggle, root MaterialApp]
    ├── theme/app_theme.dart                [Material 3 palettes, CardThemeData, and Plus Jakarta Sans text styles]
    ├── models/health_models.dart           [Core calculation engines: BMI, BMR/TDEE, Water intake, Ideal Weight]
    ├── widgets/
    │   ├── bmi_scale_bar.dart                 [4-tier color segmented BMI gauge with animated marker pin]
    │   ├── water_meter.dart                   [Circular progress hydration meter with percentage & fluid volume]
    │   ├── custom_card.dart                   [Reusable container with soft drop shadows and rounded corners]
    │   ├── unit_toggle.dart                   [Segmented pill toggle for metric and imperial measurement switches]
    │   └── disclaimer_card.dart               [Standard non-diagnostic educational health disclaimer banner]
    └── screens/
        ├── main_navigation_shell.dart         [Responsive shell with bottom navigation bar across all screens]
        ├── home_dashboard_screen.dart         [Dashboard with welcome banner, BMI quick-access, and tips preview]
        ├── bmi_calculator_screen.dart         [Interactive gender selection, age validation, cm/ft, kg/lb inputs]
        ├── bmi_result_screen.dart             [BMI score badge, visual gauge, healthy range, and actionable advice]
        ├── calculators_hub_screen.dart         [Unified TabBar host for all 4 calculators with pill tabs]
        ├── water_calculator_screen.dart       [Hydration target computation with interactive cup-by-cup logger]
        ├── calorie_calculator_screen.dart     [Mifflin-St Jeor BMR & TDEE calculator with macronutrient split]
        ├── ideal_weight_screen.dart           [Devine, Robinson, and WHO BMI target range comparison matrix]
        ├── health_tips_screen.dart            [Filterable cards with modal bottom-sheet detailed health advice]
        └── profile_screen.dart                [Ganesh (ganesh@gmail.com), streak stats, units, and dark mode toggle]
```

5. Core Implementation: Biometric Calculation Engines

The application encapsulates all medical and physical mathematics into pure, tested domain models in lib/models/health_models.dart. Below are the key mathematical formulas implemented in Dart:

Snippet 5.1: BMI & Optimal Weight Computation Engine (health_models.dart)

```
// Standard Clinical Formula: BMI = weight (kg) / [height (m)]²
static BMIResult calculate({
  required double heightCm, required double weightKg,
  required int age, required String gender,
}) {
  final heightMeters = heightCm / 100.0;
  final bmi = weightKg / (heightMeters * heightMeters);
  BMICategory category;
  if (bmi < 18.5) { category = BMICategory.underweight; }
  else if (bmi < 25.0) { category = BMICategory.normal; }
  else if (bmi < 30.0) { category = BMICategory.overweight; }
  else { category = BMICategory.obesity; }

  // WHO Normal Weight Boundary (18.5 - 24.9 kg/m²)
  final healthyMin = 18.5 * (heightMeters * heightMeters);
  final healthyMax = 24.9 * (heightMeters * heightMeters);
  return BMIResult(
    bmiValue: double.parse(bmi.toStringAsFixed(1)),
    category: category, heightCm: heightCm, weightKg: weightKg,
    healthyWeightMin: double.parse(healthyMin.toStringAsFixed(1)),
    healthyWeightMax: double.parse(healthyMax.toStringAsFixed(1)),
    age: age, gender: gender,
  );
}
```

Snippet 5.2: Mifflin-St Jeor BMR, TDEE & Macronutrient Allocation

```
// Mifflin-St Jeor: Men: 10W + 6.25H - 5A + 5 | Women: 10W + 6.25H - 5A - 161
static CalorieResult calculate({
  required double weightKg, required double heightCm, required int age,
  required String gender, required ActivityLevel activityLevel, required CalorieGoal goal,
}) {
  double bmr = (10 * weightKg) + (6.25 * heightCm) - (5 * age);
  bmr += (gender.toLowerCase() == 'male') ? 5 : -161;
  final tdee = bmr * activityLevel.multiplier;
  final target = (tdee + goal.offsetKcal).clamp(1200.0, 4500.0);

  // Balanced Macronutrient Distribution (30% Protein, 40% Carbs, 30% Fat)
  final protein = (target * 0.30 / 4).round(); // 4 kcal/gram
  final carbs = (target * 0.40 / 4).round(); // 4 kcal/gram
  final fat = (target * 0.30 / 9).round(); // 9 kcal/gram
  return CalorieResult(bmr: bmr, tdee: tdee, targetCalories: target, proteinGrams: protein, carbsGrams: carbs, fatGrams: fat);
}
```

Snippet 5.3: Clinical Ideal Body Weight Formulations (Devine & Robinson)

```
// Devine (1974) & Robinson (1983) Equations calculated for height over 60 inches
final inchesOver60 = ((heightCm / 2.54) - 60).clamp(0.0, 40.0);
double devine = (gender == 'Male') ? 50.0 + (2.3 * inchesOver60) : 45.5 + (2.3 * inchesOver60);
double robinson = (gender == 'Male') ? 52.0 + (1.9 * inchesOver60) : 49.0 + (1.7 * inchesOver60);
final whoMin = 18.5 * (heightMeters * heightMeters);
final whoMax = 24.9 * (heightMeters * heightMeters);
```

6. Core Implementation: UI Custom Components & DevOps

To deliver a modern user experience, custom interactive UI widgets were built with smooth animation and visual gauges. Automated delivery is achieved through GitHub Actions cloud runners.

Snippet 6.1: Visual Multi-Segment BMI Scale Bar with Animated Pin (bmi_scale_bar.dart)

```
// Computes precise horizontal marker position on a continuous 14.0 to 38.0 spectrum
final double minScale = 14.0; final double maxScale = 38.0;
final double normalizedRatio = (currentBmi.clamp(minScale, maxScale) - minScale) / (maxScale - minScale);

Widget build(BuildContext context) {
  return Column(children: [
    LayoutBuilder(builder: (context, constraints) {
      final pointerX = (constraints.maxWidth * normalizedRatio).clamp(16.0, constraints.maxWidth - 16.0);
      return Stack(children: [ Positioned(left: pointerX - 22, child: _buildPinBadge(currentBmi)) ]);
    }),
    ClipRRect(borderRadius: BorderRadius.circular(8), child: Row(children: [
      Expanded(flex: 45, child: Container(color: BMICategory.underweight.color)),
      Expanded(flex: 64, child: Container(color: BMICategory.normal.color)),
      Expanded(flex: 50, child: Container(color: BMICategory.overweight.color)),
      Expanded(flex: 81, child: Container(color: BMICategory.obesity.color)),
    ])),
  ]);
}
```

Snippet 6.2: Dynamic Theme Switching & Profile Management (profile_screen.dart)

```
// Configured candidate profile with instant ThemeMode (Light/Dark) toggle
Text('Ganesh', style: theme.textTheme.titleMedium?.copyWith(fontSize: 18, fontWeight: FontWeight.w800)),
Text('ganesh@gmail.com', style: theme.textTheme.bodyMedium?.copyWith(fontSize: 12)),
SwitchListTile.adaptive(
  title: const Text('Dark Mode Appearance'),
  value: widget.isDarkMode,
  onChanged: (_) => widget.onToggleTheme(), // Triggers setState() in HealthMateAppState
);
```

Snippet 6.3: Continuous Deployment Workflow (.github/workflows/deploy.yml)

```
# Automatically compiles Flutter Web release and publishes to GitHub Pages
name: Deploy HealthMate to GitHub Pages
on: [push, workflow_dispatch]
jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: subosito/flutter-action@v2
        with: { channel: 'stable', cache: true }
      - run: flutter config --enable-web && flutter pub get
      - run: |
          REPO_NAME="${{ github.event.repository.name }}"
          BASE_HREF="/$REPO_NAME/"
          flutter build web --release --base-href "$BASE_HREF"
      - uses: actions/upload-pages-artifact@v3
        with: { path: 'build/web' }
      - uses: actions/deploy-pages@v4
```

7. Application Outputs & UI Verification

The web application has been built and verified across desktop and mobile form factors. Below is a detailed tabular breakdown of each verified screen, interactive states, and calculation outputs:

Screen / View	User Inputs & Actions	Observed Screen Outputs & Dynamic Responses
Home Dashboard	App launch or tapping 'Home' tab.	Header displays 'HealthMate' with greeting; Featured 30-sec BMI CTA card; 4 Quick access tool cards with color accents; Daily Health Insights preview.
BMI Calculator	Gender: Male Age: 25 Height: 175 cm Weight: 70 kg	Interactive gender cards highlight on tap; Height supports seamless cm vs ft/in conversion; Weight toggles between kg and lb; Live boundary validations.
BMI Results Screen	Tapping 'Calculate BMI' button.	**BMI Value: 22.9 kg/m²** **Badge: NORMAL WEIGHT** (Emerald Green) Visual Gauge pointer sits inside green segment; Healthy Range: 56.7 – 76.3 kg.
Daily Water Intake	Weight: 70 kg, Moderate activity, Mild climate.	**Target Goal: 2,700 mL (11 glasses)**. Tapping '+ 1 Glass (250ml)' increments logged water; Circular progress ring animates smoothly to 46% (1,250 mL).
Calorie Needs (TDEE)	Age: 28, Male, 175 cm, 72 kg, Moderately active, Goal: Maintain.	**Basal BMR: 1,675 kcal** | **Maintenance TDEE: 2,596 kcal**. Macronutrient allocation card: Protein: 195g (30%), Carbs: 260g (40%), Healthy Fats: 87g (30%).
Ideal Body Weight	Height: 175 cm (5 ft 9 in), Gender: Male, Weight: 72 kg.	**Devine Ideal: 68.2 kg** | **Robinson Ideal: 67.0 kg** WHO Normal Spectrum: 56.7 – 76.3 kg. User advised: 'Current weight is inside optimal healthy spectrum'.
Health Insights	Tapping 'Nutrition' or 'Sleep' category filter chips.	Instantly filters cards; Tapping 'Read Guidance' opens an animated bottom-sheet modal detailing hydration and metabolic benefits.
Profile & Settings	Tapping 'Profile' tab & toggling 'Dark Mode'.	User Avatar with PRO badge; Name: **Ganesh**, Email: **ganesh@gmail.com**; Live metrics: 14 calculations, 85% hydration; Instant dark theme transition.

8. Live Cloud Deployment & Verification

Live Application URL:	https://ganeshguptha777-code.github.io/BMI/	**Hosting Platform:**	GitHub Pages (SSL Active)
Source Code Repository:	https://github.com/ganeshguptha777-code/BMI	**CI/CD Runner:**	GitHub Actions (Ubuntu-latest)
Build Status:	**BUILD & DEPLOY SUCCESS (Exit Code 0)**	**Compilation Engine:**	Flutter 3.x Web Release (WASM/JS)

9. Conclusion & Academic Learning Outcomes

The **HealthMate** project demonstrates the successful design and cloud deployment of a cross-platform Flutter mobile and web application. By leveraging **Material 3** design primitives, atomic custom widgets, and modular domain mathematical calculations, the system achieves instant responsiveness and zero-latency execution. Integrating automated CI/CD via **GitHub Actions** verified that modern web applications can be built and published entirely within cloud runners without requiring local toolchains. The project accomplishes all academic goals: accurate health modeling, responsive UI/UX engineering, and continuous cloud deployment.